

Based on the above notions, we get the existence of a N -bubble solution.

Theorem 1.1 *Suppose that (A_1) holds and either of the following holds*

- (1) $\hat{m}^* < 4$ and $\sum_{i \in \hat{I}} \sum_{t=1}^N e^{(\hat{m}^*-2)H_{i,t}(p^*)} h_i(p_t^*) D_{i,t} \neq 0$;
- (2) $\hat{m}^* = 4$ and $\sum_{i=1}^n \sum_{t=1}^N e^{2H_{i,t}(p^*)} h_i(p_t^*) L_{i,t} \neq 0$;
- (3) $\hat{m}^* = 4$, $\sum_{i=1}^n \sum_{t=1}^N e^{2H_{i,t}(p^*)} h_i(p_t^*) L_{i,t} = 0$ and $\sum_{i=1}^n \sum_{t=1}^N e^{2H_{i,t}(p^*)} h_i(p_t^*) D_{i,t} \neq 0$.

Then, there exist $\rho_\varepsilon := (\rho_{1,\varepsilon}, \dots, \rho_{n,\varepsilon})$ and a family of solutions $\{u_\varepsilon := (u_{1,\varepsilon}, \dots, u_{n,\varepsilon})\}_\varepsilon$ to Problem (1.1) corresponding to ρ_ε such that $\rho_{i,\varepsilon} \rightarrow \rho_i^*$ ($i = 1, \dots, n$) and

$$\frac{h_i(x)e^{u_{i,\varepsilon}}}{\int_\Omega h_i(x)e^{u_{i,\varepsilon}}} \rightarrow \sum_{t=1}^N 2\pi m_i^* \delta_{p_t^*} \text{ in the sense of measure}$$

as $\varepsilon \rightarrow 0+$. Here, δ_p is the Dirac measure supported at $p \in \Omega$ and the numbers $D_{i,s}$ and $L_{i,s}$ are defined as in (1.8) and (1.9). Moreover, the solution u_ε satisfies the sharp estimates (SE₁) and (SE₃). Furthermore, (SE₂) holds if Cases (1) or (2) occur, we get

$$\Lambda_{I,N}(\rho_\varepsilon) = - \sum_{i=1}^n \frac{2}{N} \left(D_{i,t} + \sum_{s \neq t} \frac{e^{2H_{i,s}(p^*)} h_i(p_s^*)}{e^{2H_{i,t}(p^*)} h_i(p_t^*)} D_{i,s} + o(1) \right) \varepsilon_t^2 \quad (1.11)$$

if Case (3) occurs.

Remark 1.2 (A_1) and Assumptions (1 – 3) in Theorem 1.1 are based on the blowup analysis of Liouville systems in [31, 23, 20]. In particular, the surprising (1.10) plays a crucial role.

Remark 1.3 The subtlety of our construction can be observed from Assumption (3) in Theorem 1.1 because it is new even for the blowup analysis in [20].

Remark 1.4 $D_{i,t}$ is a global quantity because it is involved with global integration, $L_{i,t}$ is a local quantity because its definition only needs information at blowup point. This difference is consistent with the blowup analysis not only for Liouville systems [31], but also for the singular Liouville equations [5]

Remark 1.5 Because of the difficulty on the structure of solutions, there has not been too many works on the construction of blowup solutions of Liouville systems. In [22] Huang constructed a sequence of one-bubble blowup solutions under further restrictions. In this article we removed all the extra assumptions and constructed the multi-bubble blowup solutions without additional restrictions.

Remark 1.6 Strictly speaking, the family of solutions $\{u_\varepsilon = (u_{1,\varepsilon}, \dots, u_{n,\varepsilon})\}_\varepsilon$ depends on N parameters $\varepsilon_1, \dots, \varepsilon_N$, which also depend on each other. For the sake of simplicity, we will still use ε to parameterize the solutions.